IBM Skills Build for University Engagements - AICTE-2026

Problem Statement No. 41 - Power System Fault Detection and Classification

Electrical Engineering: (Machine Learning project)

The Challenge

Design a machine learning model to detect and classify different types of faults in a power distribution system. Using electrical measurement data (e.g., voltage and current phasors), the model should be able to distinguish between normal operating conditions and various fault conditions (such as line-to-ground, line-to-line, or three-phase faults). The objective is to enable rapid and accurate fault identification, which is crucial for maintaining power grid stability and reliability.

Kaggle dataset link

https://www.kaggle.com/datasets/ziya07/power-system-faults-dataset

Technology

Use of IBM Cloud Lite services is mandatory.

Submitted by: Niranjana P